OXHACK

RoadWatch

Using historical evidence to rank which Brooklyn street blocks are most likely to receive a pothole report next week, so a fixed inspection budget is focused where it is most likely to be needed

23,448
Brooklyn street blocks to choose from

0.9%
of block-weeks get a pothole report (test period)

71,609
Brooklyn 311 pothole reports since 2020

Retrospective evaluation on public NYC and NOAA data. Predicts 311 reports, not confirmed defects. Not a live municipal deployment.

DATA COLLECTION

Real public data, with messy locations

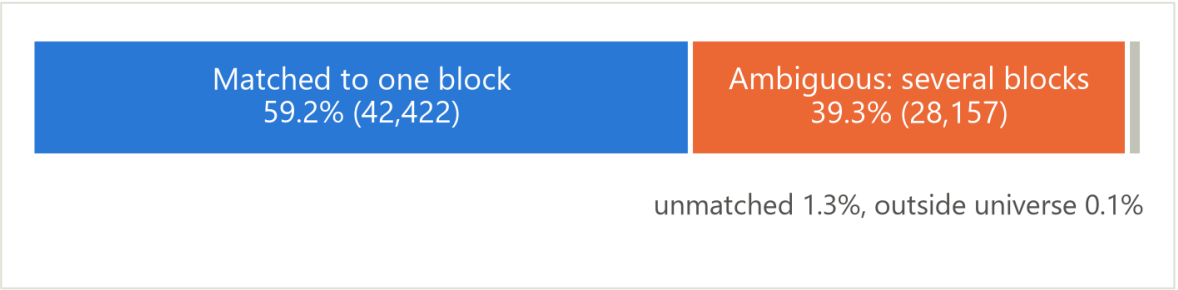
NYC 311 service requests (Socrata API)
71,609 Brooklyn pothole reports, Jan 2020 – Oct 2026

NYC street centerline (official)
23,448 eligible blocks · 2,617 km · lanes, width, truck and bike routes

NOAA GHCN-D, Central Park
daily temperature, rain and snow · 2,577 days

Also ingested for checks, not used by the model: pavement ratings (135,776 rows) and DOT work orders (110,830 rows).

Where was the pothole? Matching 311 reports to blocks



Ambiguous reports: most name an intersection (23,824 of 28,157), so a report could belong to any of its candidate blocks, typically 4.

Missing locations: 5,567 extra pothole requests had no usable borough or street; 111 were recovered with evidence.

Unknown, not zero: in 23 of 355 weeks too many reports were unlocatable, so those weeks are excluded rather than treated as 'no report'.

Source: outputs/stage3/matching_diagnostics.json, label_diagnostics.json; outputs/stage7/reproducibility.json

PROCESSING

From raw downloads to point-in-time features

1 Validate and archive

Raw API responses kept unchanged with sha256 checksums; row counts checked against the server. An offline rebuild reproduced 8/8 tables byte-for-byte.

2 Match to blocks

Street names and coordinates place each report on a block; ambiguous reports keep every candidate.

3 Label each block-week

Positive, negative or unknown (8.3 million block-weeks). Unknown weeks are never counted as zero.

4 Point-in-time features

38 features per block, built only from data published before the forecast date.

What the model is allowed to know when it ranks blocks on forecast Monday t



Source: Stage 2–4 reports; outputs/stage7/reproducibility.json; outputs/stage4/feature_definitions.md

PREDICTION

LightGBM ranks blocks, tested by replaying history

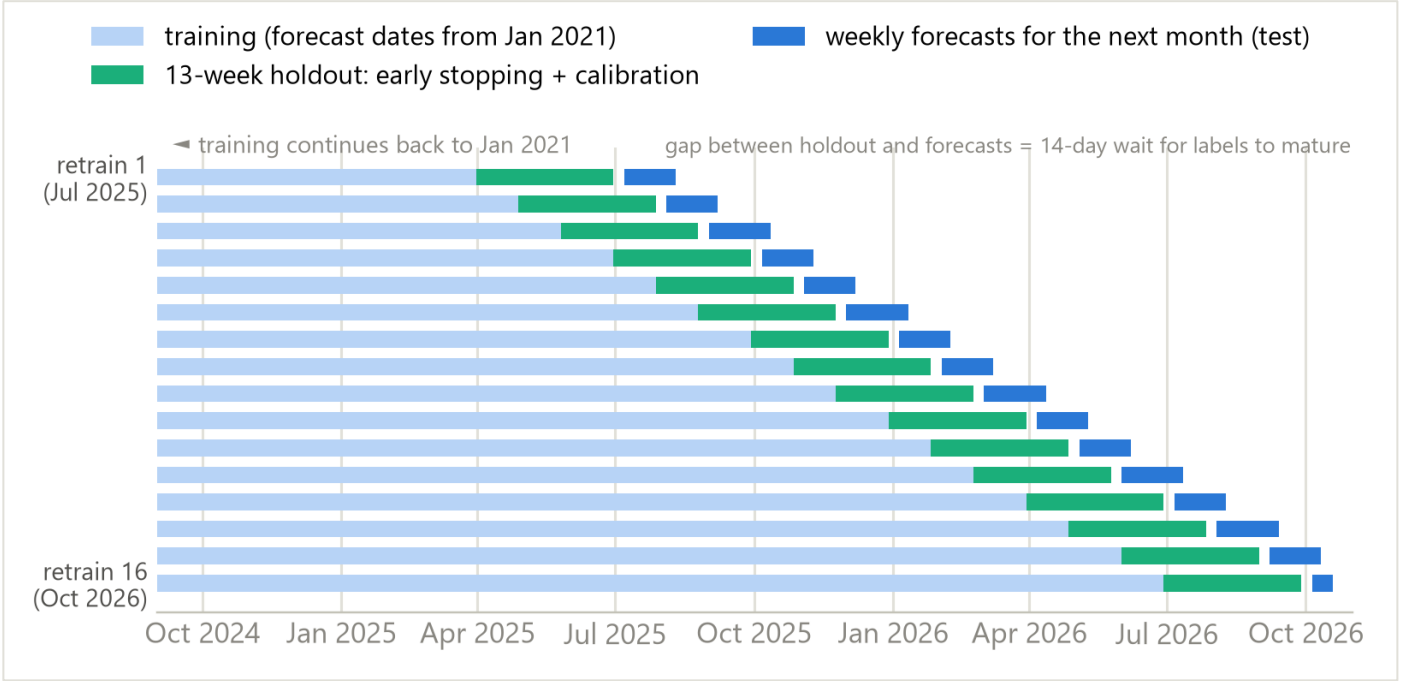
LightGBM: gradient-boosted decision trees over 38 features: recent report history (decayed counts, days since last report), neighbouring blocks, and static road attributes (length, lanes, width, truck and bike routes).

Output: a score that ranks all blocks for the next 7 days. Ranks, not reliable probabilities.

Baselines: historical rate (share of the last 52 matured weeks with a report), recency (decayed report count) and random order.

No peeking: model and features chosen on Jan–Jun 2025 validation weeks and frozen before any test prediction.

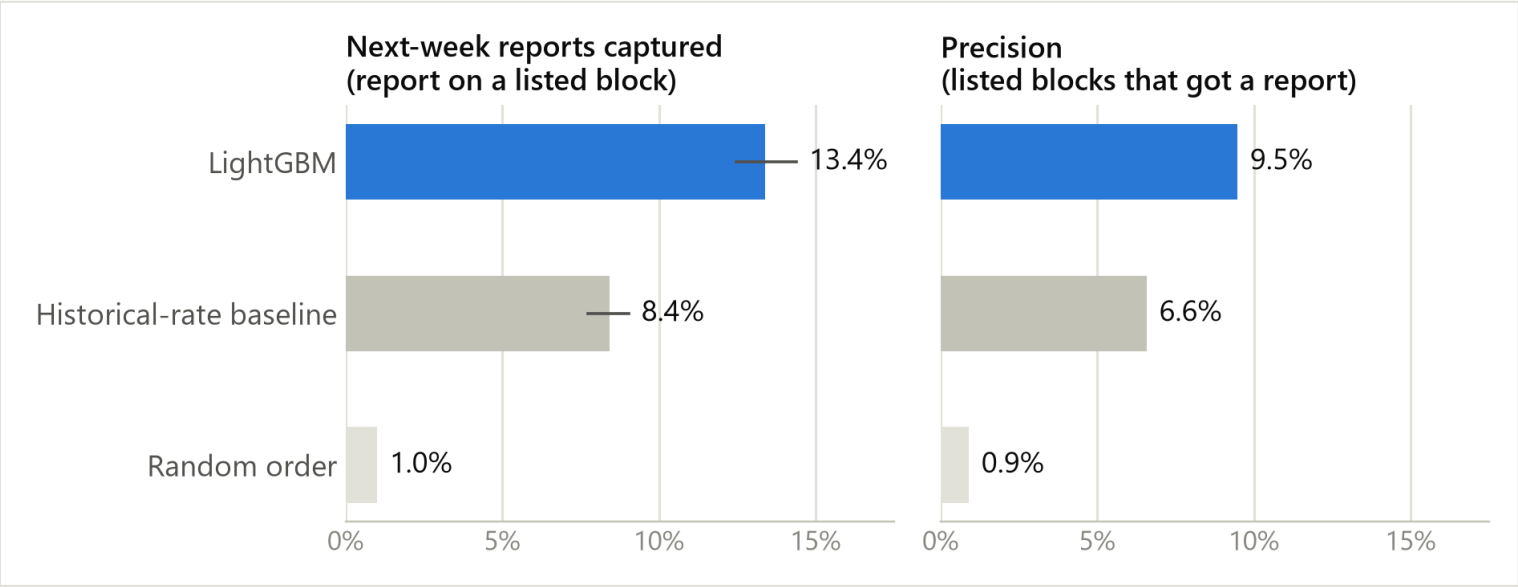
Walk-forward backtest: 16 monthly retrains, each trained only on the past



Source: data/processed/stage4/models/primary__LGBM/metadata.json; data/processed/stage4/selection.json

EVIDENCE

Strict LightGBM beats the historical-rate baseline



Top 1% list = 235 blocks per week, 67 held-out weeks (Jul 2025 – Oct 2026). Lines = 95% confidence intervals (week-level bootstrap).

Measured improvement

Precision: 9.5% vs 6.6%; paired 95% CI +2.3 to +3.4 pp.

Report capture: 13.4% vs 8.4% (random 1.0%).

Target definition matters

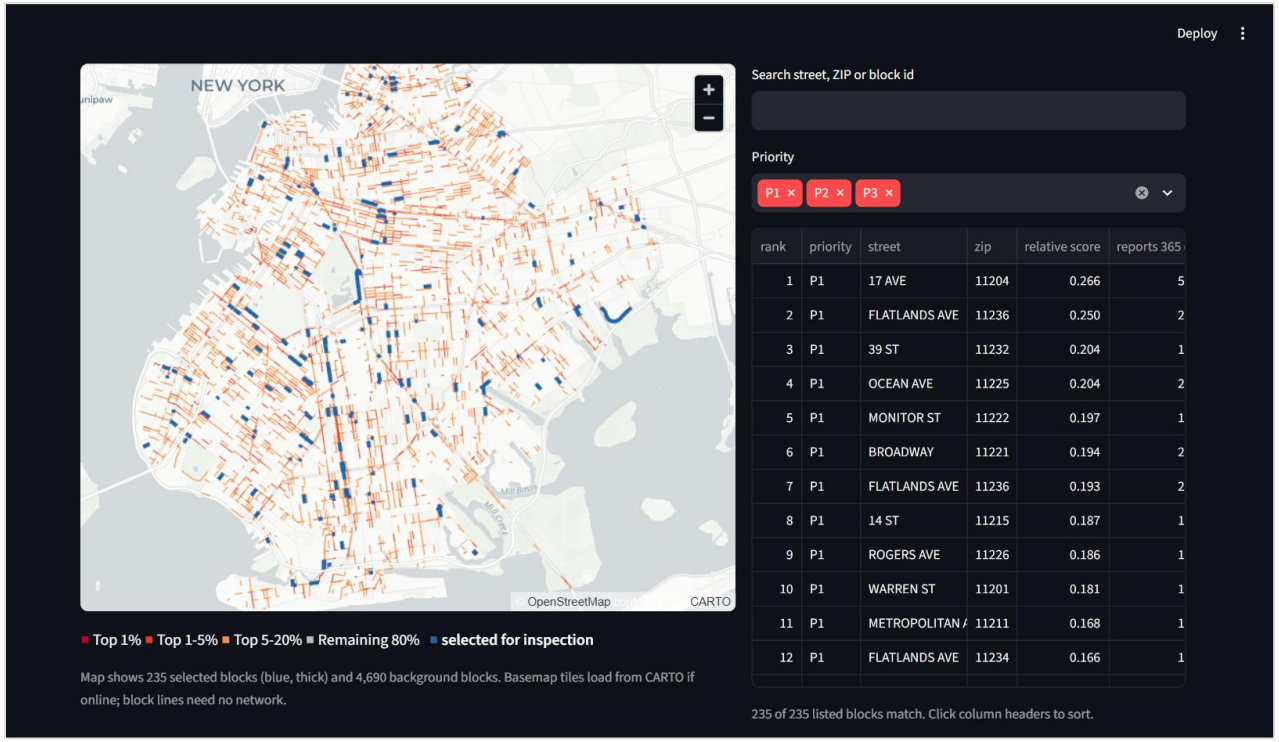
Strict single-block target: each report counts on exactly one block (ambiguous reports: a fixed pseudo-random pick).

Multi-block target: every candidate block counts. Its precision (16.4%) answers an easier question; it is not a better model.

Source: outputs/stage4/test_results.csv, bootstrap_comparisons.json; outputs/stage5/capture_definitions.csv (CIs). 67 test weeks, 16,855 reports

DECISION-MAKING

From ranks to a capacity-limited inspection worklist



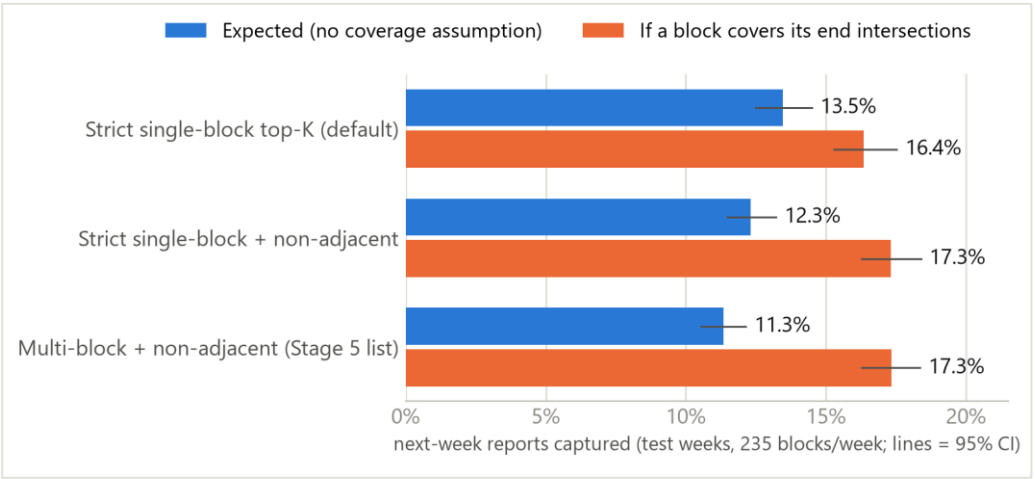
1 **Capacity:** K = 0.5% / 1% / 2% of 23,448 blocks = 118 / 235 / 469, or custom.

2 **Rank:** take the top K blocks by this week's score.

3 **Explain:** priority tier, report history, model drivers, map.

4 **Act:** inspect (not repair); export CSV or GeoJSON.

Which list is best depends on what an inspection covers

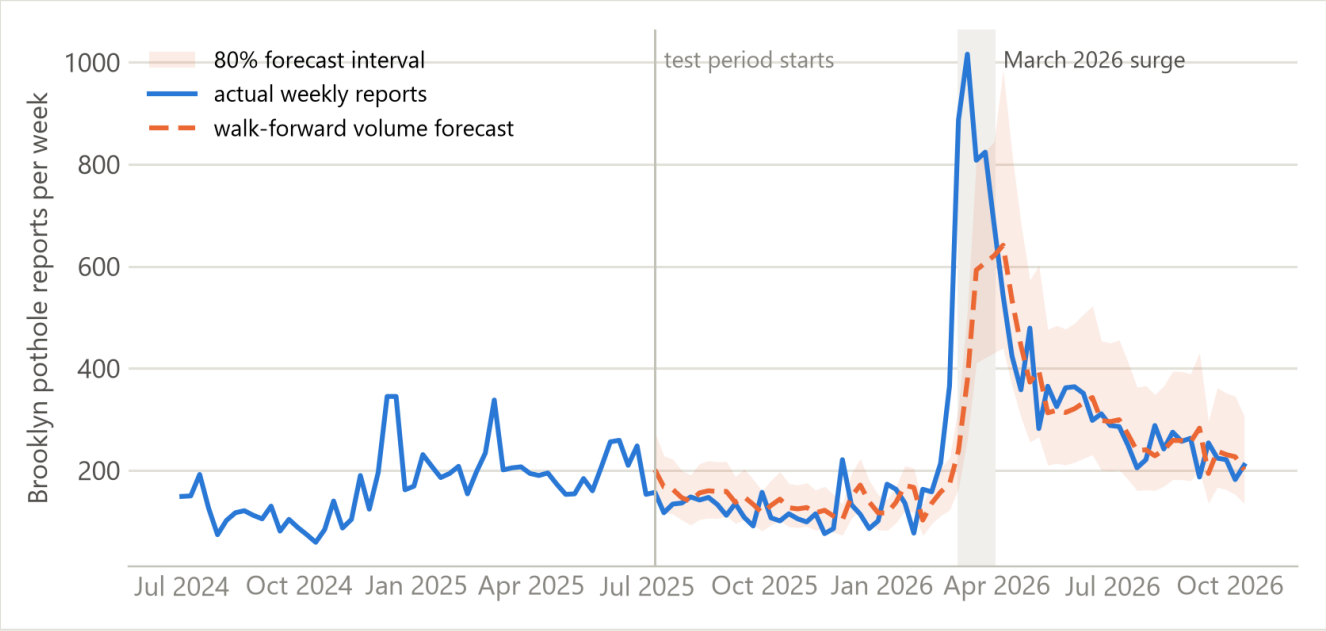


The non-adjacent alternative was proposed after test-period results were seen, so its advantage is a hypothesis, not a confirmed result.

Source: Live dashboard, retrospective week 16 Mar 2026 (default strict top-K, 1%); outputs/stage5/capture_definitions.csv

LIMITATIONS

What RoadWatch does not show (yet)



March 2026 surge: 843 reports per week, about 4× normal. The volume forecast fell 42% short; ranking still beat the baseline (12.9% vs 9.2% expected capture, only 5 weeks).

Weak volume forecasts: MAE 64.2 vs 64.1 for 'same as last week', so capacity stays a human decision.

Reports, not defects: no inspection outcomes; no failures prevented or money saved are claimed.

Reporting bias: reports track complaints; by ZIP they follow DOT work orders (rho 0.77) but not pavement ratings (rho −0.23).

Ambiguous locations: 44% of test-period reports have several candidate blocks.

Source: outputs/stage5/volume_forecast_results.json, capture_definitions.csv (March: 5 weeks), geographic_coverage.json

REAL-WORLD IMPROVEMENT

What it would take to deploy

Improvement	Data needed	How to implement	Success metric
Confirmed inspection outcomes	defect found / not found, severity, repair date	outcome columns in the exported worklist, joined back by block id each week	defects found per 100 inspections vs baseline list
Better intersection locations	GPS or photo with each 311 report	snap reports to the nearest block leg; learn which leg is usually at fault	share of reports on one block (now 59.2%)
Prospective shadow trial	future weeks after 12 Oct 2026	freeze model and default list; score weekly without tuning	pre-registered capture and precision vs baseline
Costs and repair effectiveness	repair cost, crew time, re-report rate	randomised or alternating trial against current practice	cost per verified defect; only then any savings claim

Also planned: true publication timestamps, local traffic and road-condition data, surge-aware calibration, equity monitoring, and testing weather in the volume forecast (appendix A3).

APPENDIX A1

Capture under four definitions, 1% capacity (95% CI)

Test weeks	Fractional (expected)	Assigned block	Intersection-node	Any candidate
Strict single-block top-K (default)	**13.5% (12.5–14.5)**	**13.4% (12.5–14.4)**	**16.4% (15.3–17.5)**	**17.0% (16.0–18.1)**
Strict single-block + non-adjacent	12.3% (11.5–13.2)	12.3% (11.6–13.2)	17.3% (16.3–18.4)	18.3% (17.2–19.6)
Multi-block + non-adjacent (Stage 5 list)	11.3% (10.6–12.1)	11.2% (10.4–11.9)	17.3% (16.3–18.4)	18.5% (17.4–19.7)
Plain top-K multi-block	12.8% (11.8–13.9)	12.8% (11.8–13.9)	15.5% (14.5–16.7)	16.2% (15.2–17.4)
Historical-rate baseline	8.4% (7.8–9.1)	8.4% (7.7–9.0)	10.8% (10.0–11.6)	11.5% (10.5–12.4)

Validation weeks	Fractional (expected)	Assigned block	Intersection-node	Any candidate
Strict single-block top-K (default)	**9.7% (8.4–10.9)**	**9.7% (8.4–10.9)**	**13.2% (11.7–14.7)**	**13.9% (12.5–15.4)**
Strict single-block + non-adjacent	9.0% (8.0–10.0)	8.9% (7.9–10.0)	13.8% (12.4–15.3)	14.7% (13.1–16.1)
Multi-block + non-adjacent (Stage 5 list)	8.3% (7.4–9.2)	8.1% (7.3–9.0)	13.9% (12.5–15.3)	15.0% (13.4–16.4)
Plain top-K multi-block	8.6% (7.4–9.5)	8.4% (7.3–9.5)	11.6% (10.2–12.9)	12.3% (10.7–13.6)

Source: outputs/stage5/capture_definitions.csv (Stage 7); validation = Jan–Jun 2025, test = Jul 2025 – Oct 2026

APPENDIX A2

Why the non-adjacent 'gain' depends on an assumption

Multi-block non-adjacent list minus plain top-K (reports, test, 1%)	Any candidate	Assigned block	Fractional
Reports with one candidate block	+84	+84	+84
Ambiguous intersection reports	+275	−321	−291
Ambiguous blockface / other	+30	−40	−34
Total	**+389**	**−277**	**−241**

Never listing two blocks that share an intersection raises 'any candidate' capture (+2.3 pp) because one inspection is credited with every candidate block of an intersection report. Under expected (fractional) capture the same lists lose (−1.4 pp; paired 95% CI −2.1 to −0.9 pp).

Disclosure

The non-adjacent rule was written after test-period top-K results existed. Its validation 'confirmation' was therefore not independent, and its advantage is treated as an unconfirmed hypothesis.

Default choice

Strict single-block top-K: best under expected capture and never worse than plain top-K under any definition (test, fractional +0.2 to +1.1 pp).

Source: outputs/stage7_scratch_eval/decomposition_1pct.csv; outputs/stage5/capture_definitions.csv; docs/stage7_audit_report.md

APPENDIX A3

Improvement roadmap: data, implementation and evaluation

Improvement	Data needed	Proposed implementation	Evaluation metric
Inspection outcomes and severity	found / not found, severity, repair date per inspected block	outcome columns in exported worklists; retrain on verified defects	defects per 100 inspections; severity-weighted capture
Intersection location	GPS or photo per report; inspector GPS	snap to nearest block leg; learn leg from outcomes	share uniquely located; any-vs-fractional gap
Publication timestamps	daily API snapshots (:created_at, :updated_at)	scheduled ingest; measure per-record availability lag	lag distribution; backtest with true as-of data
Traffic, weather, road condition	traffic counts, more stations, pavement ratings	add point-in-time features; validation-only selection	paired change in precision@1% and AP
Surge handling and calibration	recent weekly totals	weekly recalibration; drift alert on predicted/observed	monthly calibration ratio near 1; Brier score
Reporting equity	community-board areas, independent condition data	coverage panel; optional minimum coverage per area	inspections and defects found per area
Weather in volume forecast	NOAA daily and forecast weather	extend the tested weather feature set; walk-forward	MAE vs 'same as last week' with paired CI
Repair effectiveness and cost	repair cost, crew time, re-reports, claims	randomised or alternating trial vs current practice	cost per verified defect; time to repair

Source: Proposals only; none of these outcomes has been measured

APPENDIX A4

Reproducibility and verification

- Offline rebuild from archived raw snapshots: 8/8 processed tables, 18/18 label outputs and 6/6 feature files byte-identical.
- Frozen LightGBM refits reproduce saved predictions exactly (maximum difference 0.0).
- Leakage review: 311 lag of 2 days and NOAA lag of 5 days are exact; labels and IDs never enter a model; model selection was frozen before the first test prediction.
- Automated pytest suite; every number in this deck is re-derived from saved files by scripts/final/verify_final.py.
- Dashboard checked in a real browser: downloads byte-identical to saved worklists; invalid input handled; map works offline without tiles.

Source: outputs/stage7/reproducibility.json; outputs/final/verification/